

# Climate and Health: Verified Teaching Synthesis

Teaching Portfolio project synthesis

Verified 2026-08-09

## 1 Currentness Notice

Verified on 2026-08-09. Dataset release labels, refresh dates, and component observation periods are not interchangeable; surveillance indicators may not represent complete population incidence.

## 2 Purpose and Scope

This chapter presents a concise, source-attributed teaching synthesis of Climate and Health. It separates factual knowledge from private archival provenance and distinguishes regulations, proposals, guidance, scientific assessments, and datasets.

## 3 Evidence and Authority Framework

A defensible application identifies the issuing organization, jurisdiction, effective status, relevant date, population or geographic scope, and stated limitations before using a claim.

## 4 Current Authoritative Baseline

**Centers for Disease Control and Prevention** CDC's BRACE framework uses five iterative steps: anticipate impacts and assess vulnerabilities; project disease burden; assess interventions; develop and implement an adaptation plan; and evaluate impacts and improve activities.

**Centers for Disease Control and Prevention** CDC's climate-health resource page links the Heat & Health Tracker and Tracking Network indicators rather than serving as the underlying dataset.

**Centers for Disease Control and Prevention** CDC Tracking documents historical and projected heat, heat-related emergency visits, hospitalizations, mortality, and vulnerability/preparedness indicators from multiple sources.

**Centers for Disease Control and Prevention** Heat-related illness surveillance may miss cases when excess heat is not explicitly documented and should not be interpreted as complete population incidence.

**Centers for Disease Control and Prevention** The Heat & Health Tracker uses NSSP syndromic surveillance; its primary health source is heat-related emergency-department visits reported as rates per 100,000 ED visits.

**Agency for Toxic Substances and Disease Registry** The HHI is a national ZIP Code Tabulation Area-level index combining historical heat-health burden, sensitivity, sociodemographic, and natural/built-environment modules.

**Agency for Toxic Substances and Disease Registry** HHI uses ZCTAs to approximate ZIP-code geography and ranks indicators within four modules plus an overall rank.

**Agency for Toxic Substances and Disease Registry** The 2024 HHI covers the 48 contiguous states and District of Columbia; Alaska, Hawaii, Puerto Rico, and other U.S. territories are excluded because required indicator coverage was insufficient.

**Agency for Toxic Substances and Disease Registry** The current registered HHI data release is labeled 2024; that release label is distinct from the various observation periods of its component indicators.

**World Health Organization** WHO identifies age, health status, occupation, socioeconomic conditions, acclimatization, infrastructure, and local conditions as factors shaping heat vulnerability and impact.

**World Health Organization Regional Office for Europe** WHO/Europe's 2026 second-edition heat-health action-plan guidance is organized around eight core elements spanning governance, warning systems, vulnerable populations, communication, health-system resilience, exposure reduction, surveillance, and evaluation.

**World Health Organization** WHO describes climate-health pathways through extreme weather, heat, air and water quality, food systems, infectious disease, mental health, displacement, and health-system disruption.

**Centers for Disease Control and Prevention** CDC/ATSDR's strategic framework emphasizes health equity, surveillance and analytics, research and evaluation, preparedness/adaptation, and education/dissemination.

**U.S. Global Change Research Program** NCA5 Chapter 15 assesses interacting climate-health pathways including heat, air quality, infectious disease, water and food risks, mental health, and health-system effects.

**U.S. Global Change Research Program** NCA5 emphasizes that exposure, sensitivity, and adaptive capacity are unevenly distributed and that climate change can worsen health inequities.

**U.S. Global Change Research Program** NCA5 Chapter 30 provides region-specific evidence for Hawaii and U.S.-Affiliated Pacific Islands; findings should not be generalized uniformly across islands or communities.

## 5 Instructional Application

Learners classify source authority, test the boundaries of each claim, construct a hypothetical professional decision, and communicate what is known, proposed, recommended, or unresolved.

## 6 Common Errors

- Treating proposed regulatory text as current final law.
- Treating recommendations or guidance as automatically enforceable.
- Generalizing regional or dataset-specific findings beyond their documented scope.
- Confusing publication, release, effective, retrieval, and observation-period dates.

## 7 Limitations

This synthesis is not a legal opinion or an operational substitute for current agency text, site-specific professional judgment, or applicable state and local requirements. Public release does not establish ownership of prior academic or institutional materials.

## 8 Conclusion

Professional teaching is strengthened when factual claims are current, authority distinctions remain visible, uncertainty is explicit, and external sources are independently identifiable.

## 9 References

- U.S. Global Change Research Program. Fifth National Climate Assessment, Chapter 15: Human Health. <https://nca2023.globalchange.gov/chapter/15/>
- U.S. Global Change Research Program. Fifth National Climate Assessment, Chapter 30: Hawaii and U.S.-Affiliated Pacific Islands. <https://nca2023.globalchange.gov/chapter/30/>
- Centers for Disease Control and Prevention. Building Resilience Against Climate Effects framework. <https://www.cdc.gov/climate-health/php/brace/index.html>
- Centers for Disease Control and Prevention. Climate and Health Program data and tools. <https://www.cdc.gov/climate-health/php/resources/data-and-tools.html>
- Centers for Disease Control and Prevention. Environmental Public Health Tracking heat and climate data documentation. <https://www.cdc.gov/environmental-health-tracking/php/data-research/heat-heat-related-illness.html>
- Centers for Disease Control and Prevention. Heat and Health Tracker source documentation. <https://www.cdc.gov/nssp/php/partnerships/cdc-heat-health-tracker-uses-nssp-data.html>

- Agency for Toxic Substances and Disease Registry. Heat and Health Index landing page. <https://www.atsdr.cdc.gov/place-health/php/hhi/index.html>
- Agency for Toxic Substances and Disease Registry. Heat and Health Index technical documentation. <https://stacks.cdc.gov/view/cdc/158548>
- Agency for Toxic Substances and Disease Registry. Latest Heat and Health Index downloadable dataset and data dictionary. <https://www.atsdr.cdc.gov/place-health/php/hhi/index.html>
- World Health Organization. Heat and health fact sheet. <https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health>
- World Health Organization Regional Office for Europe. Heat-health action plans: guidance, second edition. <https://www.who.int/europe/publications/i/item/9789289062930>
- World Health Organization. Climate change and health fact sheet. <https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health>
- Centers for Disease Control and Prevention. Climate and Health Strategic Framework. <https://www.cdc.gov/climate-health/php/about/strategic-framework.html>

## Public Provenance Statement

This teaching resource is a project-generated synthesis informed by prior academic and instructional materials and updated using current authoritative sources. External sources are cited where applicable.